
16.8 SOME USEFUL REFERENCES

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Mankiw Gregory N. (2016) '*Macroeconomics*', 9th Edition, Worth Publishers, USA.

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Steven Husted and Michael Melvin (2013) '*International Economics*', Ninth Edition, Pearson Education, Inc., publishing as Addison-Wesley.

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World Bank. (1993). '*The East Asian Miracle: Economic Growth and Public Policy*'. New York: Oxford University Press.

16.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 16.2
- 2) See sub-section 16.3.2
- 3) See sub-section 16.3.3

Check Your Progress 2

- 1) See sub-section 16.4.2
- 2) See sub-section 16.4.3
- 3) See sub-section 16.5.1

UNIT 17 TRADE AND ENVIRONMENT *

Structure

- 17.0 Objective
- 17.1 Introduction
- 17.2 Trade and Environment: Linkages
 - 17.2.1 Trade and Externalities
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17.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explain how trade and environment are related to each other;
- Describe the impact of trade on the environment;
- Discuss the policies related to trade and environment;
- Explain the role of the international agencies in protecting trade and environment; and
- Describe the various agreements signed between various nations on the area of trade and environment.

17.1 INTRODUCTION

With the increase in the volume of the world population, many issues have evolved. One of them is the relationship between trade and the environment. Is trade good or bad for the environment? The answer is not that straight. There are some environmental effects when producing goods that are exported to other nations or imported from other nations. But will these effects increase or decrease with expanded trade? What is their impact on the exporting nation, the importing

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nation, or the world? And whose responsibility is it to respond to environmental hazards associated with trade? How international trading community is responding to this issue? These are some of the questions this unit will try to address.

Many emerging and developing economies often consciously adopt the export-led growth strategy in their quest for economic prosperity. A number of countries over the last two decades have achieved their economic growth by responding to the high external demand for merchandise products as well as knowledge-based services or through foreign direct investment (FDI) inflows. International trade and investment are the channels through which environmental policies transmit from one country or region to another. Although international trade is expected to harmonize environmental policies across countries in the long run, the growth process in a country may trigger sustainability challenges through various channels. The sustainability challenges may originate in various manners, namely - concentration of environmentally damaging industries (e.g., ship-breaking, toxic chemical processing), overexploitation of natural resources (e.g., surge in fishing, forestry, and mining activities), emission of greenhouse gases (GHGs) etc.

Linkages between international trade and environment are bidirectional. On one hand, stringent domestic environmental policies / regulations are often criticized for creating distortions in international trade patterns owing to the associated cost of compliance escalation and the consequent adverse implications on competitiveness. On the other hand, international trade and export liberalization policies may influence environmental sustainability through three different but integrated pathways, namely *scale effect* (attributed to accelerating economic growth), *composition effect* (attributed to structural changes in production / economy) and *technique effect* (attributed to technological progress). However, trade liberalization may not be necessarily detrimental for environment.¹

17.2 TRADE AND ENVIRONMENT: LINKAGES

First, in line with the predictions of the Environmental Kuznets Curve hypothesis (EKCH), the growing income level of a country can significantly influence its environmental sustainability. Given the availability of natural resources and labour cost advantage, foreign investment in the 'South' often extends the frontier of production and exportable surplus and brings in other concern areas. The shift in economic activities from agriculture (primary sector) to manufacturing (industrial activities) leads to large scale exploitation of both renewable (e.g., water, land, air, forest) and non-renewable (e.g., minerals, fossil fuels) natural resources, resulting in their large-scale depletion and / or

¹ For comprehensive of literature reviews on environmental impacts of international trade, see Jayadevappa and Chhatre (2000), Jeppesen *et al.* (1998), Thompson and Strohman (1996).

degradation, if adequate pollution abatement measures are not taken. The development experience of several developing countries and LDCs in Asia indicates that the industrialization drive, if not accompanied by environmental governance, often lead to sustainability challenges. Second, the Pollution Haven Hypothesis (PHH) deals with environmental degradation scenario originating from FDI inflows towards the pollution-intensive sectors / regions. Third, the Factor Endowment Hypothesis (FEH) argues that a country will specialize in the most abundantly available resource and accordingly lead to sustainability challenges. For instance, a relatively capital-abundant country may specialize in dirty capital-intensive industries, in presence of trade. However, the PHH and FEH work in opposite directions, as specialization patterns for rich (capital-abundant) and poor (labour-abundant) countries differs accordingly. Finally, over-reliance on a natural resource may lead to policy myopia and compromise the broad-based growth path, resulting in the Natural Resource Curse Hypothesis (NRCH) phenomenon. While the existence of NRCH in itself may not necessarily involve environmental challenges, specialization in 'brown' activities, particularly in mining or quarrying industries, may lead to serious concerns. These effects, either in isolation or cumulatively, often lead to not only local (e.g., release of chemical effluents) environmental problems but also trans-boundary (either regional or global) concerns (e.g., climate change concerns).

The nature of public goods of environmental problems results in market failure. It demands government interventions to set the property rights correctly and / or adopt economic policies to internalize the environmental costs.² However, inter-jurisdiction competitions to attract investment and industries, by providing economic incentives in addition to weak environmental rules and regulations and/ or lenient enforcement of the rules, often result in 'race to the bottom' competition. In other words, the lax national environmental standard in a country might function as an implicit subsidy to the local players (vis-à-vis foreign players from a country characterized by stringent environmental standards) in the sense that the marginal cost of abatement would remain below the marginal damage from pollution. However, if the export partners of a country enhance their demand for quality (e.g., through adoption of ISO 14001 standards), such trade linkages may then nullify the 'race to the bottom' forces. It has been argued that adoption of a strong environmental standard may be an optimal policy for governments, where the marginal cost of abatement would exceed the marginal environmental damage. Apart from indirect environment policy-led incentives,

² Non-inclusion of environmental costs in private marginal cost often results in shifting of environmental incidence (impacts / problems) to a large section of the society as they have to bear the costs in terms of adoption of environmental averting practices (e.g., water purifier, air purifier) and / or costs of morbidity and mortality due to environmental pollution. Therefore, the centralised private costs (the cost of pollution abatement) are shifted to public at large who may not get any benefits from the economic activity that generates pollution.

tax incentives-cum-subsidies are often used to attract investments which are not always conducive for sustainability scenario.

Since the basic trade theory does not consider environmental externalities associated with the production and/ or consumption of goods, following Harris and Roach (2017) a graphical approach of trade and environment is presented below to familiarize underlying basic concepts of trade with environmental externalities.

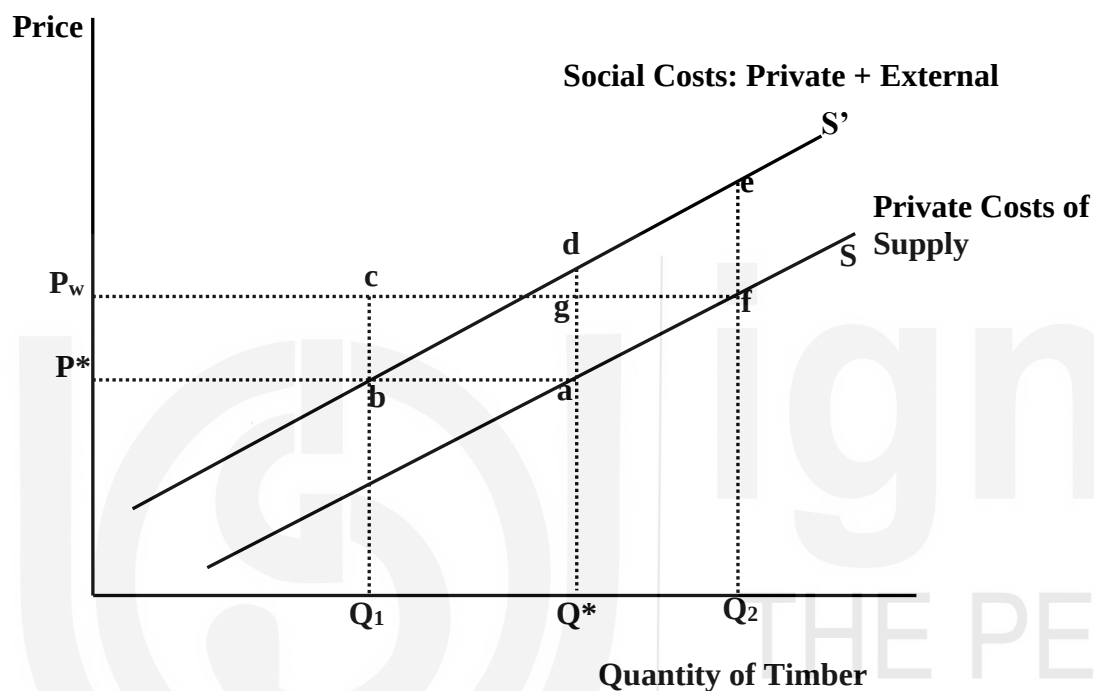


Figure 17.1 Gains and Losses from Importing Automobiles

In the absence of international trade, domestic supply and demand will be in equilibrium at 'a' and corresponding equilibrium output will be Q^* and domestic price will be P^* . If there are no barriers to trade, automobiles can be imported at the world price P_w . Domestic production will fall to Q_1 , as domestic automobile manufacturers will lose market share due to availability of cheaper imported automobiles. Domestic consumption will increase to Q_2 , as more consumers will find affordable to purchase imported automobiles. The quantity of import will be $Q_2 - Q_1$.

The domestic automobile manufacturers will lose an area P^*abP_w as they sell fewer units and that also in lower price. Domestic manufacturers will also incur loss of output equivalent to $Q^* - Q_1$. Domestic consumers will gain an area of P^*acP_w as price of automobiles fall and purchase more units. The net gain from trade will be area abc .

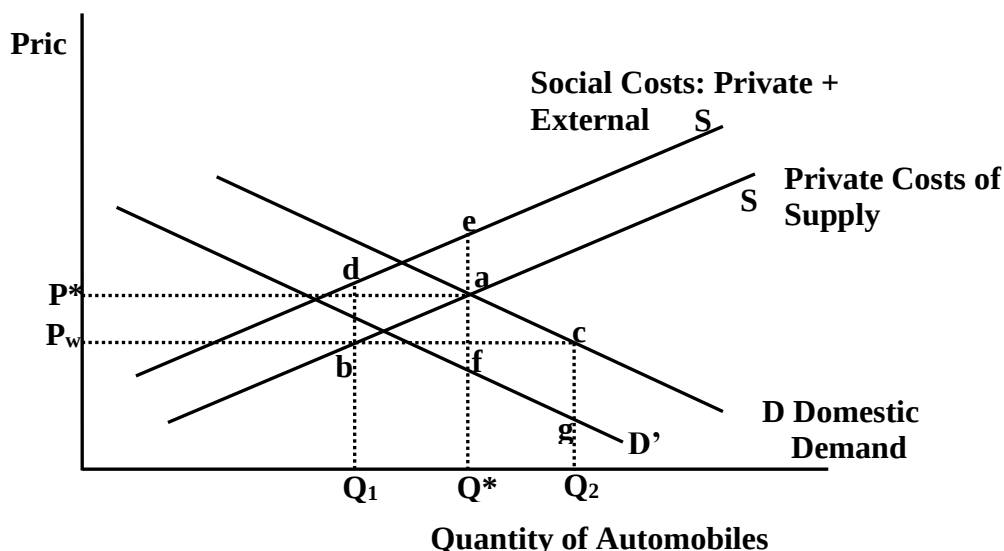


Figure 17.2: Environmental Impacts of Importing Automobiles

17.2.1 Trade and Externalities

However, this trade is not free from environmental externalities. There are two types of externalities may arise due to such trade - a) production externalities caused by production of / manufacturing of automobiles and b) consumption of externalities associated with usage of automobiles. Therefore, the gain from trade reduces if we include such externalities. Manufacturers do not include the environmental costs of production in their private cost function as a result they pass on the costs to the society in terms of air and water pollution and dumping of industrial wastes on land and water bodies. If one includes the costs associated with pollution abatement and resource recovery, costs of production will increase and the new supply curve is shown as S' in Figure 17.2. Similarly, consumers do not take into consideration the environmental externalities associated with usage of the automobile as a result it causes air pollution. If consumers are not charged for the production of externalities (exhaust emission), they pass on the costs to the society in terms of air pollution and related costs of morbidity and mortality. Therefore, benefit that accrues to consumers is reduced, as they need to incur costs to avoid exposure from pollution and/ or investing in pollution control devices (e.g., catalytic converter). Therefore, the new demand curve shifts downward at D' .

The resulting welfare impacts from trade would be area $abde$ and $afgc$ in Figure 17.2, where reduction in domestic production will result in saving in environmental costs (or environmental space) of area $abde$. In other words, the country will shift these costs to exporting country(ies) by importing cars. By importing more automobiles, the country will incur additional environmental costs of the area $afgc$ in terms of air pollution. Since, relative size of the areas ($abde$ and $afgc$) may vary depending on the environmental impacts. Therefore, it would be difficult to draw conclusions unequivocally whether trade will result in net gain or loss for a country. Moreover, it will depend on goods/commodities

involved in trade. Importing those goods whose manufacturing is detrimental for environment and there is no consumption related environmental impacts will benefit importing countries. On the other hand, importing goods whose consumption leads to environmental impacts and there is no production related environmental implication will benefit exporting countries.

To clarify consumption and production related environmental externalities and their impacts on international trade, we describe them individually below.

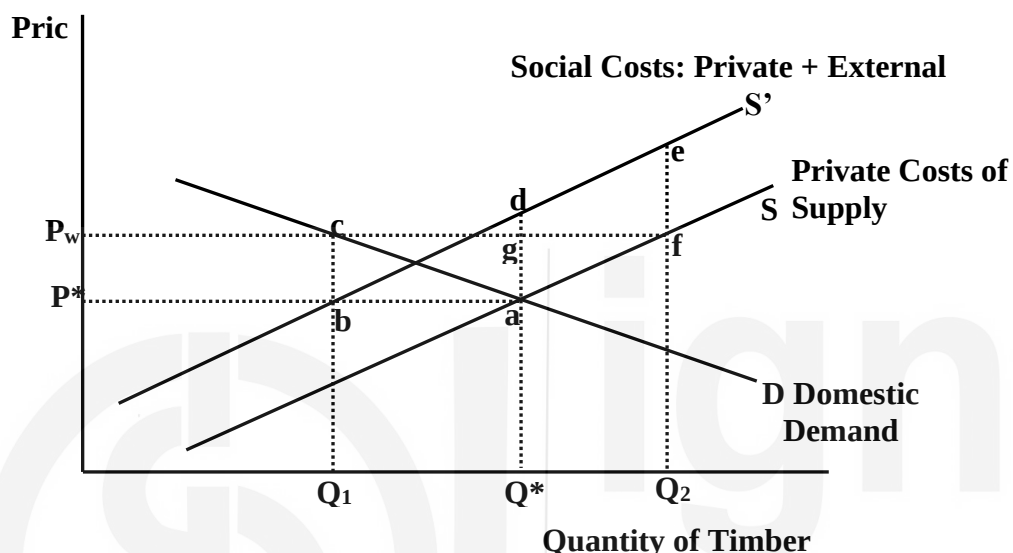


Figure 17.3: Production Externalities and International Trade

Exports of natural resources like timber and minerals result in environmental costs associated with development (extraction) of natural resources. Higher price of such natural resources in international market may lead to over-exploitation and therefore may cause large scale deforestation and rapid drawing down of mineral resources. Therefore, social costs of such natural resources are higher than private costs. Since domestic consumers have to pay higher price (P_w), they will lose consumer surplus by the area P_wcbP^* and producers of such natural resources will gain by the area P_wfaP^* . The additional gains of the producers are supported by higher price (P_w) as well as additional production of natural resources ($Q_2 - Q^*$). Additional social cost associated with such export will be the area equivalent to $adef$. There may have some social costs (externalities) associated with consumption (processing) of such natural resources in the importing countries, however for simplicity we have not considered them here. The gain from the trade - equivalent to the area $abcf$ will be weighed against additional social cost of the export - equivalent to the area $adef$. Therefore, the ultimate gain or loss from the trade will depend on relative size of the areas $abcf$ and $adef$. This example is also applicable for exports of all commodities where environmental externalities arise in the production (e.g., manufacturing textiles, leather, and metals).

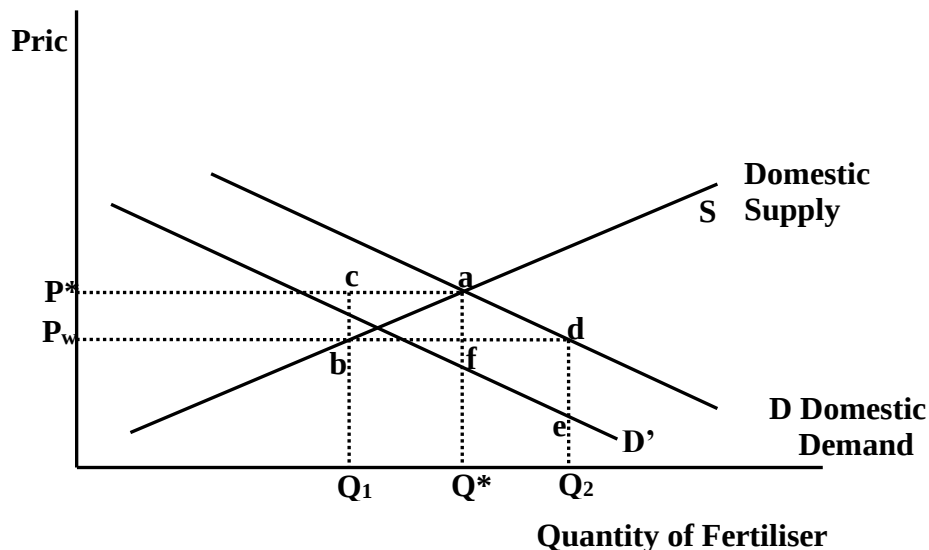


Figure 17.4: Consumption Externalities and International Trade

Import of hazardous wastes (say electronic waste) for processing and disposal is an example where severe environmental externalities are associated with demand or consumption. In the absence of international market for e-waste, domestic recyclers will pay a price P^* to procure the waste from domestic suppliers (say aggregators / waste handlers) and recycle up to Q^* (Figure 17.4). If international e-waste is available at a price of P_w ($< P^*$), domestic recyclers will import e-waste and process up to Q_2 ($> Q^*$). The gains of domestic recyclers will be the area adP_wP^* , where domestic aggregators will lose by the area bP_wP^*c . However, recycling of such wastes is associated with substantial environmental externalities as well as health costs to recyclers (due to exposure to heavy metals and poisonous gases). Therefore, the benefits from recycling are lower and costs associated with the externalities will be the area equivalent to $adef$. Therefore, ultimate benefits or loss associated with such type of e-waste trade will depend on relative size of the areas $adbc$ and $adef$. However, it is to be kept in mind that unless government intervenes and impose an environmental tax/ fee on such recyclers, they will succeed to retain the benefits of additional recycling of wastes and pass on the costs to the society in terms of environmental and public health impacts.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Explain the relationship or linkages between International Trade and Environment.

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